

Text Retrieval and Search Engines



Lecture 7 – Query Expansion



Some slides adapted from the companion website to the "Introduction to IR" book by Manning, Raghavan and Schutz & Victor Lavrenko



The Vocabulary Mismatch Problem

- Vocabulary mismatch between terms appear in short queries and relevant documents
 - A core challenge in ad hoc document retrieval
 - For high recall – E.g., searching for *baby* doesn't match with *infant*; nor *car* with *automobile*
- Inter-document similarities
- Semantic similarities
- Relevance feedback
- Query expansion
- ...

Synonymy and Polysemy

- Vocabulary mismatch can be caused by:
 - Polysemy – a single term with multiple meanings
 - table, count, tank, bank
 - Synonymy – a single meaning can be expressed by different terms
 - car <-> automobile, baby <-> infant
- Natural language is ambiguous
- User should have sufficient knowledge to be able to make an initial query
 - User may be unfamiliar with the domain (e.g., medical)
 - May not have full command of the language (non-native)

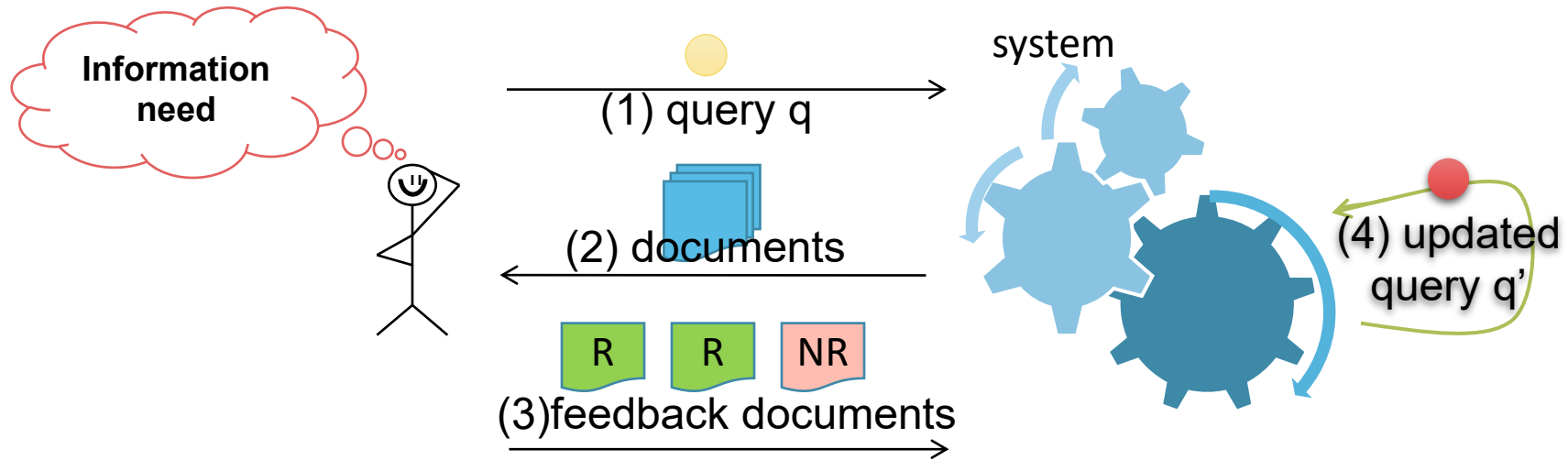
How to Improve Recall?

- Global methods
 - Query expansion/reformulation
 - Thesauri (e.g., Wordnet)
 - Automatic thesaurus generation
- Local methods
 - Relevance feedback
 - Pseudo relevance feedback

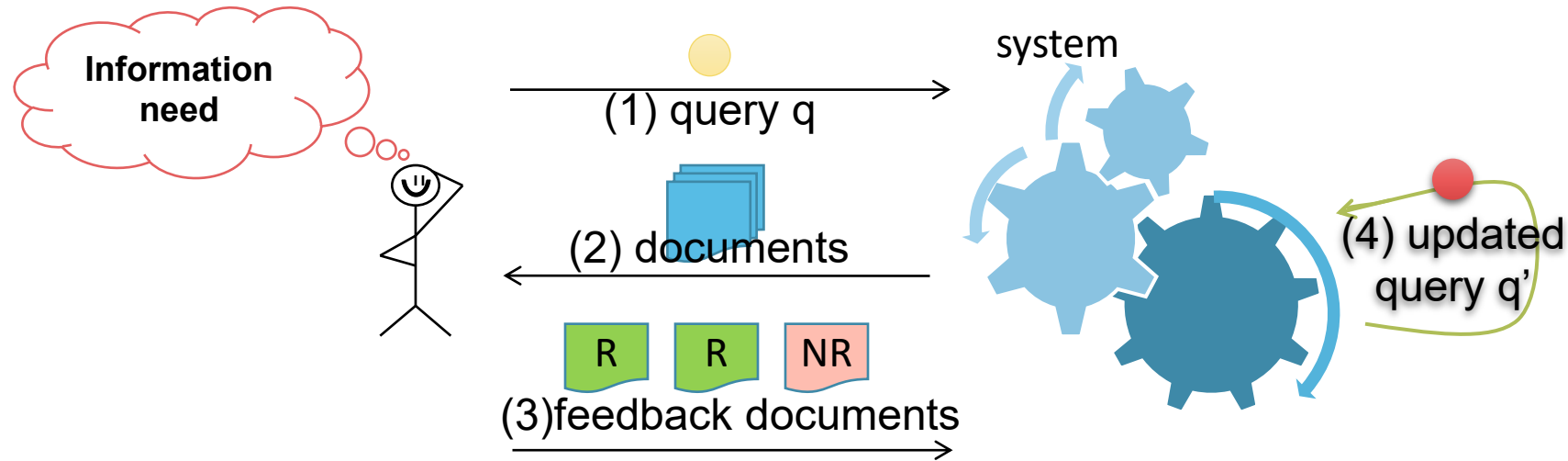


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Relevance Feedback



Relevance Feedback



- Allows the system to update the original query representation
 - May be iterative
 - May be implicit (e.g., from clicks)
- Idea: it may be difficult to formulate a good query when you don't know the collection well, so iterate

Rocchio's Algorithm (SMART)

[Rocchio '71]

- Rocchio's algorithm incorporates relevance feedback information into the vector space model
- Want to maximize $\text{sim}(q, D_r) - \text{sim}(q, D_{nr})$
- The optimal query vector for separating relevant and non-relevant documents (with cosine sim.)

Rocchio's Algorithm (SMART)

[Rocchio '71]

$$\vec{q}_m = \alpha \vec{q}_0 + \beta \frac{1}{|D_r|} \sum_{\vec{d}_j \in D_r} \vec{d}_j - \gamma \frac{1}{|D_{nr}|} \sum_{\vec{d}_j \in D_{nr}} \vec{d}_j$$

Centroid of relevant documents

Centroid of non-relevant documents

- q_m = modified query vector; q_0 = original query vector
- D_r = set of known relevant doc vectors; D_{nr} = set of known irrelevant doc vectors
- α, β, γ – tune the importance of relevant/non-relevant terms

Rocchio's Algorithm (SMART)

[Rocchio '71]

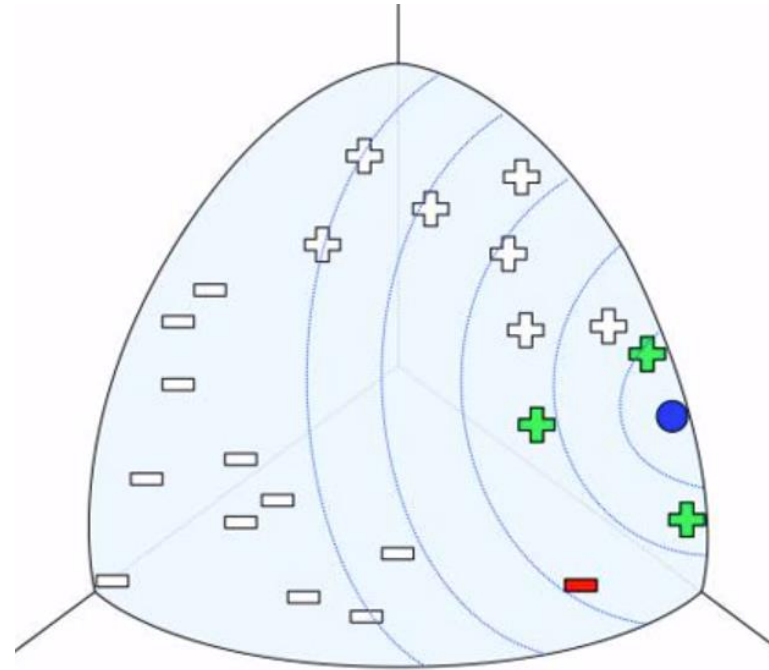
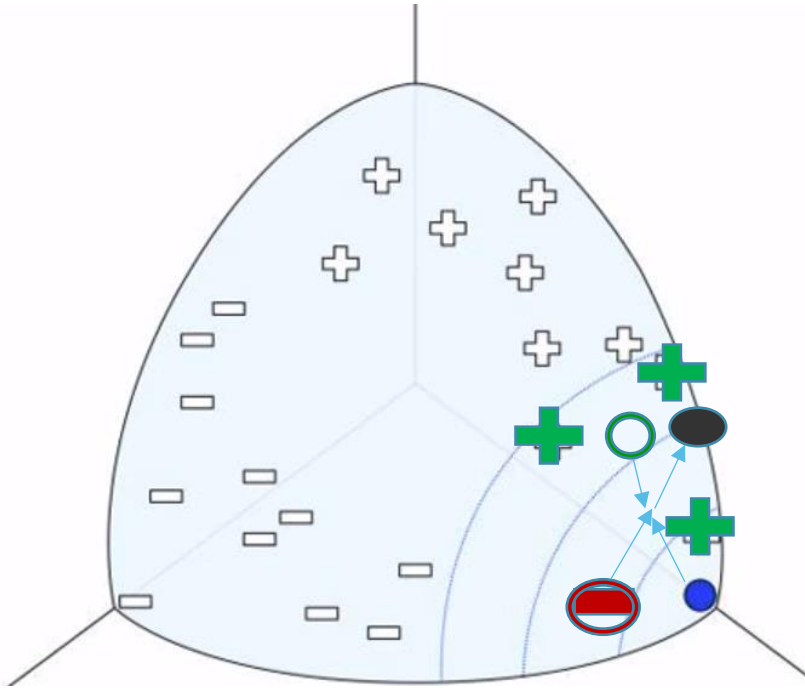
$$\vec{q}_m = \alpha \vec{q}_0 + \beta \frac{1}{|D_r|} \sum_{\vec{d}_j \in D_r} \vec{d}_j - \gamma \frac{1}{|D_{nr}|} \sum_{\vec{d}_j \in D_{nr}} \vec{d}_j$$

Centroid of relevant documents

Centroid of non-relevant documents

- New query moves toward relevant documents and away from irrelevant documents
- Tradeoff α vs. $\frac{\beta}{\gamma}$: if we have a lot of judged documents, we want a higher $\frac{\beta}{\gamma}$
- Term weight can go negative
 - Negative term weights are ignored (set to 0)
 - Doesn't make sense in the vector space model

Example



Example

- Given the following vectors and parameters, calculate the resulting feedback query
 - Relevant documents:
 - D1: (2, 3, 0, 1, 5)
 - D2: (6, 1, 4, 1, 1)
 - Irrelevant documents:
 - DN1: (2, 2, 3, 0, 4)
 - Original query:
 - q0: (0, 3, 1, 2, 2)
 - Constants:
 - $\alpha=1, \beta=0.75, \gamma=0.25$

Relevance Feedback in Vector Spaces

- We can modify the query based on relevance feedback and apply standard vector space model
- Use only the docs that were marked
- Relevance feedback can improve recall and precision
- Relevance feedback is most useful for increasing *recall* in situations where recall is important
 - Users can be expected to review results and to take time to iterate

Positive vs. Negative Feedback

- Positive feedback is more valuable than negative feedback
- Many systems only allow positive feedback -> why?

Relevance Feedback: Assumptions

- A1: User has sufficient knowledge for initial query
- A2: Relevance prototypes are “well-behaved”
 - Term distribution in relevant documents will be similar
- Term distribution in non-relevant documents will be different from those in relevant documents
 - Either: all relevant documents are tightly clustered around a single prototype
 - Or: there are different prototypes, but they have significant vocabulary overlap
 - Similarities between relevant and irrelevant documents are small

Violation of A1

- User does not have sufficient initial knowledge
- Examples:
 - Misspellings (Brittany Speers)
 - Cross-language information retrieval
 - Mismatch of searcher's vocabulary vs. collection vocabulary
 - Baby/infant

Violation of A2

- There are several relevance prototypes
 - Pop starts that worked in Burger King

Relevance Feedback: Problems

- Long queries are inefficient for typical IR engine
 - Long response times for user
 - High cost for retrieval system
 - Partial solution
 - Only reweight certain prominent terms
 - Perhaps top 20 by term frequency
 - Users are often reluctant to provide explicit feedback
 - It's often harder to understand why a particular document was retrieved after applying relevance feedback

Poison Pills

- Documents that are marked as relevant, yet when used alone for feedback yield reduced retrieval effectiveness
- Relevant documents may contain unrelated terms that do not represent the set of relevant documents



Relevance Feedback: Evaluation

- Use q_0 and compute precision-recall graph
- Use q_m and compute precision-recall graph
- In most cases q_m is spectacularly better than q_0 !
 - **Is this a fair evaluation?**
- Empirically, one round of relevance feedback is often very useful. Two rounds is sometimes marginally useful

Other Uses of Relevance Feedback

EI

what is information retrieval



Information retrieval (IR) is the process of retrieving relevant information from a collection of documents or data sources. The goal of IR is to help users find the information they are looking for, typically through a search engine or other retrieval system. 👍👎

The process of IR involves several stages, including indexing, querying, and ranking. In the indexing stage, the content of each document is analyzed and represented in a structured format that can be easily searched. In the querying stage, the user specifies their search query, which is typically a set of keywords or phrases. The retrieval system then searches the indexed documents for those that match the query, using algorithms that take into account factors such as the relevance of the document to the query and the quality of the document itself. In the ranking stage, the retrieved documents are ranked in order of their relevance to the query, and presented to the user.

IR has many applications, including search engines, recommendation systems, and data mining. It is used in a wide range of fields, from business and finance to medicine and scientific research.



Relevance Feedback Summary

- Relevance feedback has been shown to be very effective at improving relevance of results
- Requires queries for which the set of relevant documents is medium to large
- Full relevance feedback is painful for the user
- Full relevance feedback is not very efficient in most IR systems

Pseudo Relevance Feedback

- Automates the “manual” part of true relevance feedback
- Retrieve an initial set of most relevant documents
- *Assume* that the top K ranked documents are relevant
- Do relevance feedback

- Mostly works
 - Danger of query drift
 - Query is about “copper mines” and the top retrieved documents are all about mines in Chile, then there may be query drift in the direction of documents on Chile

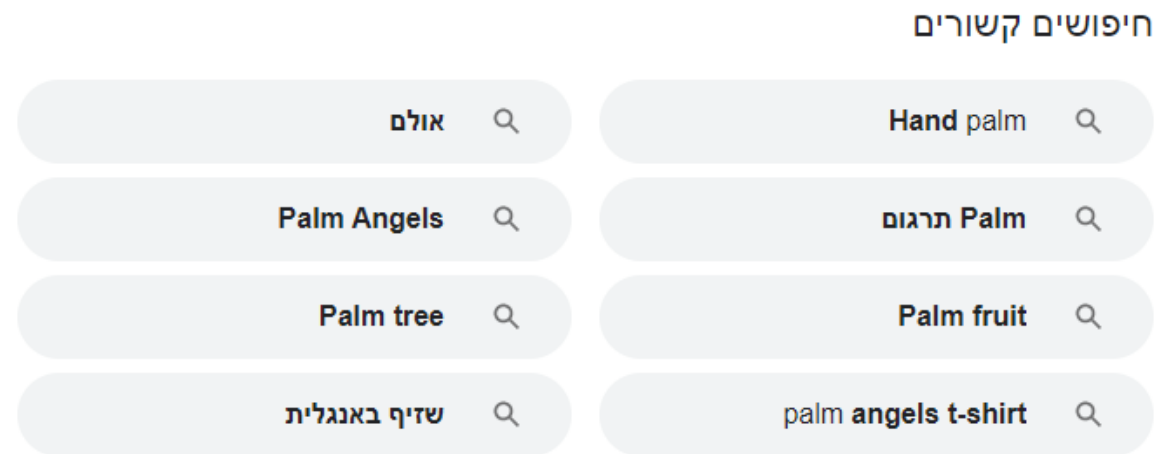
Pseudo Relevance Feedback: Cornell SMART at TREC 4

- Precision@50:
 - Inc.ltc (no PsRF) 64.2%
 - Inc.ltc (with PsRF) 72.7%
 - Lnu.ltu (no PsRF) 74.2%
 - Lnu.ltu (with PsRF) 87%

PsRF: adding 20 terms to original query

Query Expansion

- In relevance feedback, users give additional input (relevant/non-relevant) on documents, which is used to reweight terms in the documents
- In query expansion, users give additional input (good/bad search terms) on words or phrases
 - Another method for increasing recall
- Global methods
 - Manual thesaurus
 - Wordnet, MeSH
 - Automatic thesaurus generation
 - Co-occurrence statistics
 - Refinements based on query log mining
 - Common on the Web



Synonyms: MeSH Thesaurus

- Used for the MedLine database – indexed all medical research studies
- Different research made by different researchers which use different terminology to refer the same things
 - Excellent results in **exact** domains
 - Errors in unrestricted search
 - Searching in Google “Neck Pain” using “Cervical Pain”
 - Development: time and resource intensive

MeSH Heading	Neck Pain
Tree Number	C10.597.617.576
Tree Number	C23.888.592.612.553
Tree Number	C23.888.646.501
Entry Term	Cervical Pain
Entry Term	Neckache
Entry Term	Anterior Cervical Pain
Entry Term	Anterior Neck Pain
Entry Term	Cervicalgia
Entry Term	Cervicodynia
Entry Term	Neck Ache
Entry Term	Posterior Cervical Pain
Entry Term	Posterior Neck Pain

WordNet

- Generic and high-quality resource
- No notion of context – WordNet is designed for expanding single terms
- Out-of-vocabulary meanings

S: (n) **squid**, [calamari](#), [calamary](#) ((Italian cuisine) squid prepared as food)

- [domain category](#)
- [direct hypernym](#) / [inherited hypernym](#) / [sister term](#)
 - S: (n) [seafood](#) (edible fish (broadly including freshwater fish) or shellfish or roe etc)
- [domain region](#)
- [part holonym](#)

S: (n) **squid** (widely distributed fast-moving ten-armed cephalopod mollusk having a long tapered body with triangular tail fins)

- [direct hyponym](#) / [full hyponym](#)
 - S: (n) [loligo](#) (somewhat flattened cylindrical squid)
 - S: (n) [ommastrephes](#) (extremely active cylindrical squid with short strong arms and large rhombic terminal fins)
 - S: (n) [architeuthis](#), [giant squid](#) (largest mollusk known about but never seen (to 60 feet long))
- [part meronym](#)
- [direct hypernym](#) / [inherited hypernym](#) / [sister term](#)
 - S: (n) [decapod](#) (cephalopods having eight short tentacles plus two long ones)
 - [direct hyponym](#) / [full hyponym](#)
 - S: (n) **squid** (widely distributed fast-moving ten-armed cephalopod mollusk having a long tapered body with triangular tail fins)
 - S: (n) [cuttlefish](#), [cuttle](#) (ten-armed oval-bodied cephalopod with narrow fins as long as the body and a large calcareous internal shell)
 - S: (n) [spirula](#), [Spirula peronii](#) (a small tropical cephalopod of the genus *Spirula* having prominent eyes and short arms and a many-chambered shell coiled in a flat spiral)

Thesaurus-Based Query Expansion

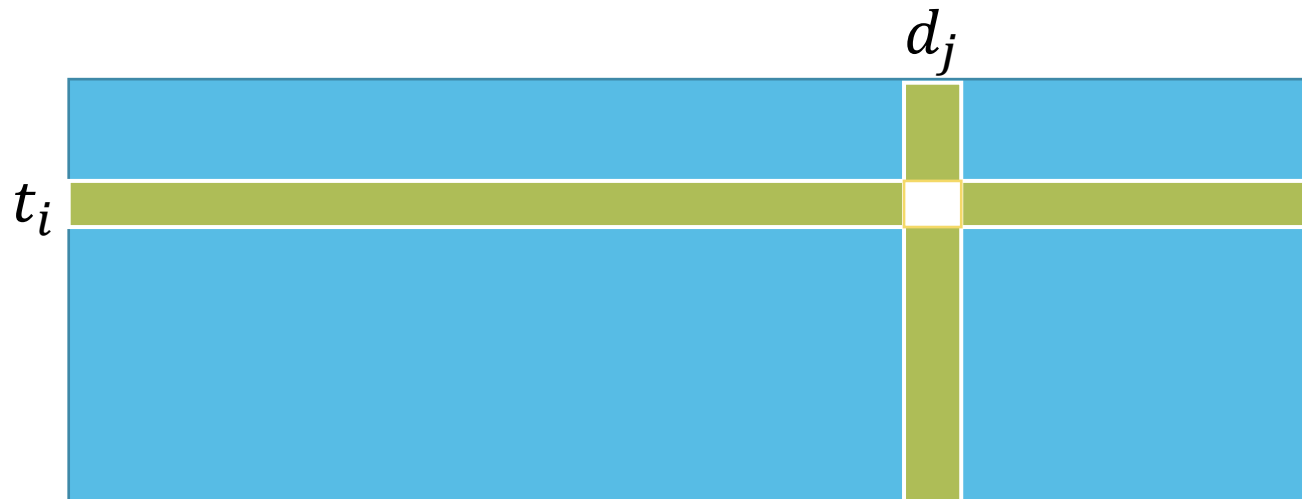
- For each term in the query, expand the query with words the thesaurus lists as semantically related with the term
- Generally increases recall
- May weight added terms less than original query terms
- Widely used in many science and engineering fields
- May significantly decrease precision, particularly with ambiguous terms
- It is very expensive to create a manual thesaurus and to maintain it over time

Automatic Thesaurus Generation

- Attempt to generate a thesaurus automatically by analyzing the collection of documents
- Two main approaches
 - Co-occurrence based (co-occurring words are more likely to be similar)
 - Shallow analysis of grammatical relations
 - You can harvest, peel, eat, prepare, apples and pears, so apples and pears must be similar

Co-occurrence Thesaurus

- Simplest way to compute one is based on term-term similarities
- Use a term-document matrix A
 - Each cell $A_{t,d}$ is a weighted count $w_{t,d}$ for term t in document d
- Calculate $C = AA^T$
 - $C_{u,v}$ is a similarity score between terms u and v



Automatic Thesaurus Generation Discussion

- Term ambiguity may introduce irrelevant correlated terms
 - Apple computer -> Apple red fruit computer
- Problems
 - False positives: Words deemed similar that are not
 - False negatives: Words deemed dissimilar that are similar
- Since terms are highly correlated anyway, expansion may not retrieve many additional relevant documents

Query Expansion Summary

- Query expansion is often effective in increasing recall
 - Not always good with general thesauri
- In most cases, precision is decreased, often significantly -> query drift
- Overall, not as useful as relevance feedback; may be as good as pseudo-relevance feedback
 - Much more understandable to the user